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CS/IT-1771
Examination- Nov-Dec - 2018
B.E. VII Sem: CSE / IT
Artificial Intelligence

Time: 3 Hrs

Max. Marks: 70
Min. Marks: 22

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:
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Word limit NOT to be followed for diagram, numerical, derivation.

1. (a) Write the name of any four characteristics of problems that required the AI techniques for solution. 02
   (b) Write the heuristic function of DFS and BFS. 02
   (c) Write down the all possible states of water jug problem. 03
   (d) Differentiate 07
   (i) Monotonic and non mono tonic production system.
   (ii) Write down the production system of water jug problem.
   (iii) Write the algorithm of generate and test algorithms.

OR

   (e) Consider trying to solve 8 puzzle using hill climbing. Can you find a heuristic function that makes it work? Make sure it works on following example 07

1	2	3
8	5	6
4	7	

1	2	3
4	5	6
7	8	

2. (a) Write the name of any four Knowledge representation techniques. 02
   (b) Write the difference between predicate and propositional logic. 02
   (c) A Problem Solving search can process either forward (from a known start state to a desired goal state) or backward (from a goal state to start state). What factors determine the choice of direction for a particular problem? 03
   (d) Consider following facts 07
   A. The members of Elm St. Bridge Club are Joe, Sally, Bill, and Ellen'
   B. Joe is married to Sally
   C. Bill is Ellen's brother
   D. Spouse of every married person in club in also member of club
   E. The last meeting of club was at Joe's

   I. Represent facts in predicate logic

II. Decide on truth values of following statements

i) Last meeting of club was at Sally's house

ii) Ellen is not married

OR

(e) Write the steps of resolution with example?

3. (a) Which information is useful crisp or fuzzy? Justify your answer.
   (b) State Bayesian theorem .
   (c) Write any three features of semantic maps.
   (d) What is planning? Explain the components of Planning.

OR

(e) Suppose we want to use semantic net to discover relationships that could help in disambiguating the word "bank" in the sentence

John went downtown to deposit his money in the bank
The Financial institution meaning for the bank should be preferred over the river bank meaning

a. Construct a Semantic net that contains representations for the relevant concepts
b. Show how intersections search could be used to find the connections between the correct meaning for the bank and the rest of the sentence more easily than it can find a connection with the incorrect meaning

4. (a) Draw the structure of Artificial Neuron.
   (b) Write the different phase of NLP.
   (c) Compare the Biological and Artificial Neuron.
   (d) Explain perception learning algorithm for training a neural network. What are the limitations of this algorithm?

OR

(e) Explain the structure and working of Hopfield net .

5. (a) Write the names of any for languages that can be used to implement AI Techniques.
   (b) Define LISP AI Programming Language in brief ?
   (c) Write the name of any six statements of Prolog.
   (d) Write the characteristics of programming language that can be used to implement the AI techniques.

OR

(e) Write a Prolog program that verifies whether an input list is a palindrome.
Hint: Goal: palindrome([r,a,c,e,c,a,r])
Output: Yes
Goal: palindrome([a,b,c])
Output: No

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CS/IT-1771
Examination – Nov/Dec.- 2019
B.E. VII Sem : CSE / IT
Artificial Intelligence

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1.	(a)	What is greedy best first search?	02
	(b)	When A* is optimal?	02
	(c)	What are the factors that affect the quality of a heuristic?	03
	(d)	Explain constraint satisfaction problem with an example.	07
		OR	
		Explain Hill climbing in detail with example.	07
2.	(a)	What are the components of a first order logic?	02
	(b)	What is the difference between the two quantifiers in the logics?	02
	(c)	What are the various resolution strategies?	03
	(d)	Give the Syntax and Semantics of a first order logic in detail with an example.	07
		OR	
		Explain forward chaining and backward chaining in detail for a first order definite clauses.	07
3.	(a)	Write the components of fuzzy logic.	02
	(b)	Write the name of methods of membership value assignment.	02
	(c)	Write the difference between fuzzification and defuzzification?	03
	(d)	State Bays Theorems? Write the application of Bays theorems ?	07
		OR	
		Explain the various defuzzification methods with example.	07

4.	(a)	List the activation function of neural network.	02
	(b)	What are the two choices for activation function?	02
	(c)	Write the limitation of neural network.	03
	(d)	What is ANN ? Write the names of ANN learning algorithms.	07
		OR	
		What is recurrent network ? Write the advantages of recurrent network.	07
5.	(a)	Write the name of statements of Prolog.	02
	(b)	Write one example to represent the predicate in Prolog.	02
	(c)	Explain the recursion in LISP.	03
	(d)	Discuss the features of LISP.	07
		OR	
		Discuss the features of Prolog.	07

Examination – Dec- 2023
B.Tech. VII Sem : Computer Science & Engineering
Artificial Intelligence

Time : 3 Hrs

Max. Marks : 70
Min. Marks : 22

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			Marks	BL	CO
Q.1	(a)	What are the types of Artificial Intelligence?	02	2	1
	(b)	What is the history of Artificial Intelligence?	02	1	2
	(c)	Explain the structure of agents in Artificial Intelligence.	03	2	1
	(d)	Write the name of any characteristics of problems that required the AI techniques for solution.	07	2	2
		OR			
		Explain Artificial Intelligence Techniques.	07	2	1
Q.2	(a)	What is Knowledge Representation in Artificial Intelligence?	02	2	3
	(b)	Explain Hill Climbing.	02	2	1
	(c)	What are the types of knowledge in Knowledge Representation?	03	2	2
	(d)	Write the heuristic function for best first search. State when best first search become depth first search.	07	2	3
		OR			
		Write the step of resolution with example.	07	2	2
Q.3	(a)	Explain Certainty Factor in AI.	02	2	2
	(b)	Which information is useful in crisp or fuzzy? Justify your answer.	02	2	2
	(c)	Difference between Monotonic Reasoning Vs Non-monotonic Reasoning.	03	2	1

Examination—Dec-2023

B.Tech. VII Sem : Computer Science & Engineering

Artificial Intelligence

Time : 3 Hrs | Max. Marks : 70 | Min. Marks : 22

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	(d)	*Explain Reasoning in Artificial Intelligence and its type.*	07	2	2
		OR			
		What is Bayesian reasoning? What does a Bayesian network represents? Explain.	07	2	1
Q.4	(a)	What is Continuous Planning?	02	2	3
	(b)	What is statistical learning? Definition and examples?	02	2	2
	(c)	What is Reinforcement Learning?	03	2	2
	(d)	Describe Non-Linear Planning with an example and Constraints Posting in Non-Linear Planning.	07	1	1
		OR			
		What is the Role of Planning in Artificial Intelligence? Also explain Forward State Space Planning (FSSP) and Backward State Space Planning (BSSP).	07	2	3
Q.5	(a)	Difference between Recursion and Iteration.	02	2	1
	(b)	What is Knowledge Acquisition?	02	1	2
	(c)	List few types of Predicates in LISP.	03	2	1
	(d)	Describe Prolog – Lists, its representation and Basic Operations on Lists.	07	2	2
		OR			
		Define LISP function, syntax and its usage.	07	2	2

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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CS – 1874 (A)

Examination –May- 2024

B.Tech. VII Sem: Computer Science & Engineering

Artificial Intelligence

Time : 3 Hrs

Max. Marks : 70
Min. Marks : 22

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:
Part a – Max 50 words, Part b – Max 50 words,
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

Q.	Part	Question	Marks	BL	CO
Q.1	(a)	What is the role of intelligent agent?	02	1	1
	(b)	Which technique was developed to determine whether a machine could or could not demonstrate the artificial intelligence.	02	2	1
	(c)	What are the characteristics of an intelligent agent?	03	2	1
	(d)	What is meant by the term AI? How is it different from the natural intelligence?	07	2	1
		OR			
		What is Artificial Intelligence Technique"? Briefly explain how AI Technique can be represented and list out some of the task domain of AI.	07	2	1
Q.2	(a)	Describe the first order logic model.	02	2	2
	(b)	In how many ways knowledge can be represented.?	02	2	2
	(c)	Differentiate between knowledge representation & knowledge acquisition.	03	2	2
	(d)	Give the outline of hill climbing algorithm.	07	2	2
		OR			
		Explain the A*search strategies.	07	2	2
Q.3	(a)	What is the purpose of weak slot and filler structures.	02	3	3
	(b)	What is the Baye's rule?	02	2	3
	(c)	Write brief note on Fuzzy Logic.	03	2	3
	(d)	What do you understand by the term uncertainty in AI? How is it managed?	07	2	3
		OR			
		What is non – monotonic reasoning? Explain the logics used for non – monotonic reasoning.	07	3	3

Q.4	(a)	The Bayesian Network gives a complete description of domain. Comment.	02	2	
	(b)	Write brief note on multi agent planning.	02	2	4
	(c)	What are the advantages and disadvantages of nonlinear planning?	03	2	4
	(d)	What do you mean by learning? Explain briefly learning methods.	07	2	4
		OR			
		What are the components of the planning? Explain each of them in brief.	07	2	4
Q.5	(a)	What is lisp? Why is it so popular in AI?	02	2	5
	(b)	Write the square root function in lisp.	02	3	5
	(c)	Demonstrate with an example how you can code in LISP?	03	3	5
	(d)	Explain how prolog programming language is good for robot programming.	07	4	5
		OR			
		What are the features of the prolog language.	07	2	5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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Total Printed Pages: 02

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CS-1874/1771

Examination –Nov- 2022

B.Tech. VII Sem: Computer Science & Engineering

Artificial Intelligence

Time:3 Hrs

Max. Marks:70
Min. Marks:22

Note:Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.

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Q.1 (a) Illustrate components to define a problem that required AI Techniques.? Define them. 02

(b) List down the characteristics of intelligent agent? 02

(c) List down the characteristics of problem that required AI Techniques to solve the problem. 03

(d) Show at-least two points of difference between: 07

(i) Systems that think like human and systems that acts like human.

(ii) Systems that think rationally and systems that acts rationally

OR

Define Artificial Intelligence and list the task domains of Artificial Intelligence. 07

Q.2 (a) List the two advantages of knowledge represented as logic. 02

(b) Determine WFF for following statement. 02

(i) Marchus was man. Marcus was pompion.

(ii) All pompions are roman.

(c) Enumerate the criteria to choose the search algorithms. 03

(d) Differentiate Breadth First Search & Depth First Search on the following two points. 07
(i) Time complexity (ii) Space complexity.

OR

Draw the first three levels of Breadth First Search & Depth First Search tree of the water jug problem. 07

Q.3 (a) Compare and contrast Forward vs Backward reasoning. 02

(b) State bays theorem. 02

(c) Distinguish between crisp sets in fuzzy logic 03

(d) Describe the membership function in fuzzy logic. Also summarize interference in fuzzy logic? 07

OR

State Bay's Theorem. Determine innovative example of Bayesian network. 07

Q.4 (a) List the components of Planning. 02

(b) Enumerate the types of Planning. 02

(c) Determine the real word example of reinforcement learning. 03

(d) Define Machine Learning? Mention any five machine Learning techniques. 07

OR

Summarize the components of Planning with example. 07

Q.5 (a) Mention any three programming languages for AI.

(b) Distinguish between iteration and recursion. 02

(c) List the Knowledge acquisition techniques. 03

(d) Formulate in Prolog predicates for following parent relation: 07

(a) Pat'sparent

(b) Liz has achild

(c) Pat'sgrandparent

OR

Make a program for Printing String, Atom and List in LISP. 07

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